

Median Index Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/median

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Abstract

Median selection as unique strict k -th order statistic. Pairwise gaps $> 2\varepsilon$ preserve the index under $\|\delta\|_\infty \leq \varepsilon$. Lean: `LEAN_FULL`.

1 Theorems

Theorem 1 (Preservation). *Unique strict k -th smallest with all pairwise gaps $> 2\varepsilon$ is preserved under $\|\delta\|_\infty \leq \varepsilon$.*

Theorem 2 (Sharpness). *A rival gap $\leq 2\varepsilon$ admits a tying adversary destroying uniqueness.*

2 Lean / certificate

`Research.Operators.Median.Preservation`; `LEAN_FULL` Mathlib $\mathbb{R}$.

3 Limitations

Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).